Chapter 3

Binomial Theorem

Only One Option Correct Type Questions

1. For $r = 0, 1, \ldots, 10$, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1 + x)^{10}$, $(1 + x)^{20}$ and $(1 + x)^{30}$. Then $\sum_{r=1}^{10} A_r\left(B_{10}B_r - C_{10}A_r\right)$ is equal to **[IIT-JEE-2010 (Paper-2)]**

   (A) $B_{10} - C_{10}$ (B) $A_{10}\left(B_{10}^2 - C_{10}A_{10}\right)$

   (C) 0 (D) $C_{10} - B_{10}$

2. Coefficient of x^{11} in the expansion of $(1 + x^2)^4(1 + x^3)^7(1 + x^4)^{12}$ is **[JEE (Adv)-2014 (Paper-2)]**

   (A) 1051 (B) 1106

   (C) 1113 (D) 1120

One or More Option(s) Correct Type Questions

3. For nonnegative integers s and r, let

$$\binom{s}{r} = \begin{cases} \dfrac{s!}{r!(s-r)!} & \text{if } r \le s, \\ 0 & \text{if } r > s. \end{cases}$$

For positive integers m and n, let

$$g(m,n) = \sum_{p=0}^{m+n} \frac{f(m,n,p)}{\binom{n+p}{p}}$$

where for any nonnegative integer p,

$$f(m,n,p) = \sum_{i=0}^{p} \binom{m}{i}\binom{n+i}{p}\binom{p+n}{p-i},$$

Then which of the following statements is/are TRUE? **[JEE (Adv)-2020 (Paper-2)]**

(A) $g(m, n) = g(n, m)$ for all positive integers m, n

(B) $g(m, n + 1) = g(m + 1, n)$ for all positive integers m, n

(C) $g(2m, 2n) = 2g(m, n)$ for all positive integers m, n

(D) $g(2m, 2n) = (g(m, n))^2$ for all positive integers m, n

Integer / Numerical Value Type Questions

4. The coefficients of three consecutive terms of $(1 + x)^{n+5}$ are in the ratio 5 : 10 : 14. Then n =

   [JEE (Adv)-2013 (Paper-1)]

5. The coefficient of x^9 in the expansion of $(1 + x)(1 + x^2)(1 + x^3) \ldots (1 + x^{100})$ is **[JEE (Adv)-2015 (Paper-2)]**

6. Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1 + x)^2 + (1 + x)^3 + ... + (1 + x)^{49} + (1 + mx)^{50}$ is $(3n + 1)\ {}^{51}C_3$ for some positive integer n. Then the value of n is

[JEE (Adv)-2016 (Paper-1)]

7. Let $X = ({}^{10}C_1)^2 + 2({}^{10}C_2)^2 + 3({}^{10}C_3)^2 + ... + 10({}^{10}C_{10})^2$, where ${}^{10}C_r$, $r \in \{1, 2, ..., 10\}$ denote binomial coefficients. Then, the value of $\frac{1}{1430}X$ is _____.

[JEE (Adv)-2018 (Paper-2)]

8. Suppose $\det\begin{bmatrix} \sum_{k=0}^{n} k & \sum_{k=0}^{n} {}^nC_k k^2 \\ \sum_{k=0}^{n} {}^nC_k k & \sum_{k=0}^{n} {}^nC_k 3^k \end{bmatrix} = 0$ holds for some positive integer n. Then $\sum_{k=0}^{n} \frac{{}^nC_k}{k+1}$ equals___.

[JEE (Adv)-2019 (Paper-2)]

❑ ❑ ❑